Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/03/2013
PI Name:	Poirier, David Sc.D.		
Project Title:	Effect of Varying Convection on Dendrite Morphology and Macrosegregation		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721-0012	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	12/01/2010	**End Date:**	11/30/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2014 per NSSC information (Ed., 9/2/13)		
Key Personnel Changes/Previous PI:	Professor Jhon Jairo Olaya Florez of the Universidad Nacional de Colombia, Bogata spent part of his sabbatical with the research team at The University of Arizona during February-August of 2013.		
COI Name (Institution):	Erdmann, Robert (University of Arizona) Tewari, Surendra (Cleveland State University)		
Grant/Contract No.:	NNX10AV40G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research is ground-based research on directional solidification (DS) of metallic alloys. The investigators are David Poirier (DP) and Robert Erdmann of The University of Arizona and Surendra Tewari (ST) of Cleveland State University. DP and ST have collaborated for many years, and the requisite facilities and team are in place. Hence, the research is cost-effective and promises to achieve "flight status" in minimal time. The research of this grant pertains to convection that is driven by DS through changes in cross-sections (because of solidification shrinkage) and thermosolutal convection. In microgravity, of course, the thermosolutal convection will be mitigated so that effects of the shrinkage-driven flow in cross-sectional changes can be isolated from the usual thermosolutal convection that always occurs when solidifcation occurs on earth. The outcome of the research is expected to be relevant to commercial DS-castings, that are used in the high-temperature sections of gas-turbine engines (i.e., jet engines and electric power generators). Such sophisticated castings have geometries, which invariably have changes in cross-sections. Yet convection at a cross-sectional change and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals have not been studied well. Hence, the research not only has scientific merit, it is also relevant to the production of commercial DS castings. The proposed ground-based research comprises the following: 1) directional solidification experiments that are compatible with the solidification cartridge assemblies (SCAs) that can be employed in a furnace on the International Space Station ISS. The research comprises: 1) selected alloys have varying density-responses to gravity forces; 2) crucible designs with variable cross-sections but compatible with the current SCAs; 3) sophisticated characterizations of the solidified microstructures, including both micro- and macro-segregation; and 4) supporting computer simulations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The outcome of the research is expected to be relevant to commercial directionally solidified-castings, that are used in the high-temperature sections of gas-turbine engines (i.e., jet engines and turbine electric power-generators). Such sophisticated castings have geometries, which invariably have changes in cross-sections. Yet convection at a cross-sectional change and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals have not been studied well. Hence, the research is very much needed for its scientific merit and also because it is relevant to the production of commercial DS castings.
Task Progress:	The research involves the directional solidification (DS) of Al-Si and Al-Cu hypereutectic alloys in graphite molds. The molds are designed such that DS occurs in a uniform cross-section and also through an abrupt change of area. Both contractions and expansions are studied. In mold-lengths with a uniform cross-section, the alloys exhibit "steepling." Steepling is a form of macrosegregation in which the periphery of the sample is enriched in solute; typically this enrichment shows up as the eutectic. At a change in cross-section, there is also macrosegregation. The macrosgregation is largely the result of thermosolutal convection in 1-g. At a change in cross-section, the convection is more complex because continuity requires an additional component of flow to "feed" the solidification shrinkage of the alloy. Our experimental results show that the extent of both types of macrosegregation is somewhat more in the Al-Cu alloy than it is in the Al-Si alloy because Cu increase the density of liquid-Al more than Si increases the density. We have also performed computer simulations to observe the convection and the connection between the convection patterns and the macrosegregation. Above the expected macrosegregation, the simulations have been very instructive because they have revealed solidification patterns wherein the liquid in advance of the solidifying mushy zone is undercooled. This fits nicely with our experimental observations that grain multiplication occurs at a change in cross-section (especially at an expansion). We have recently quantified (by image analyses) the radial macrosegregation in the samples of both alloys. These will results will be reported in the next progress report. We believe that this study of convection and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals has profound scientific merit and it is relevant to the production of commercial DS castings. During this reporting period, a report of progress was made at a panel review at MSFC in March, 2013. The panel's recommendations for future work are being pursued, starting with the submission of a Supplement (submitted July, 2013). Included in the Supplement are an initial design of experiments and strategy for planning possible 0-g experiments.
Bibliography Type:	Description: (Last Updated: 06/06/2025)
Abstracts for Journals and Proceedings	Lauer MA, Johnson L, Poirier DR, Tewari SN, Erdmann RG. "Simulations of the Effects of Mold Properties on Directional Solidification." To be presented at the ASME 2013 International Mechanical Engineering Congress and Exposition, Computational Modeling of Microstructural Evolution, San Diego, CA, November 15-21, 2013. ASME 2013 International Mechanical Engineering Congress and Exposition, Computational Modeling of Microstructural Evolution, San Diego, CA, November 15-21, 2013. Proceedings to be published. , Nov-2013